

11. Where does the multiplier operand is stored? 1 1 3 1
 A) Instruction register B) Shift register
 C) Cache D) Program counter
12. How many bits are needed to represent floating point number in a single precision form? 1 1 3 1
 A) 16 B) 50
 C) 32 D) 64
13. Which register in the processor is single-directional? 1 2 4 2
 A) MDR B) PC
 C) MAR D) Temp
14. The Program Counter gets incremented 1 2 4 2
 A) After the instruction decoding B) After the fetch cycle
 C) After the IR instruction gets executed D) After the read cycle executed
15. In multiple Bus organization, the registers are collectively placed and referred to as 1 2 4 1
 A) Register file B) Set registers
 C) Register Block D) Map registers
16. _____ memory is used in pipelining to increase the memory access speed. 1 2 4 2
 A) Special memory locations B) Special purpose register
 C) Cache D) Buffers
17. _____ gives the measure of how many operations can be performed in parallel at the same time. 1 2 5 1
 A) Data parallelism B) Task level parallelism
 C) Instruction level parallelism D) Transaction level parallelism
18. SIMD stands for 1 1 5 1
 A) Simple Instruction Multiple Decoding B) Single Instruction Multiple Data
 C) Sequential Instruction Multiple Decoding D) System Information Mutable Data
19. Processor which is complex and expensive to produce is 1 1 5 2
 A) RISC B) EPIC
 C) Multi-core D) CISC
20. The cost of a parallel processing is primarily determined by 1 2 5 2
 A) time complexity B) switching complexity
 C) circuit complexity D) power complexity

PART - B (5 x 8 = 40 Marks)
 Answer ALL the Questions

- 21 a. Draw the graphic symbol, algebraic function and truth table for the following logic gates. 8 2 1 1
 i. AND
 ii. OR
 iii. NAND
 iv. EX-OR

(OR)

- b. i) Perform BCD Addition for 175 and 326. (4 marks) 8 2 1 1

- ii) Perform (111100) - (001011) using 1's and 2's complement method. (4 marks)

- 22 a. Explain in detail about the different types of bus structures connected between CPU and Memory. 8 2 2 2

(OR)

- b. Discuss the Addressing modes with suitable examples. 8 2 2 2

- 23 a. Divide 10 / 3 using Non-Restoring Method. Show the operations performed in each step. 8 3 3 1

(OR)

- b. i). What are the advantages of bit pair recoding of multipliers? 8 3 3 1

- ii). Multiply (+13)x(-6) using bit pair recoding method.

- 24 a. Draw the Single Bus Organization and explain the execution of an instruction. 8 2 4 1

(OR)

- b. List and briefly explain the classes of Hazards that degrades the performance of Pipeline. 8 2 4 1

- 25 a. Write about the cache coherence problem and give solutions for this problem. 8 2 5 2

(OR)

- b. Discuss on the features, advantages and disadvantages of ARM Processor. 8 2 5 2

PART - C (1 x 15 = 15 Marks)
 Answer ANY ONE Questions

Marks BL CO PO

- 26 Describe the procedure of constructing a fast adder and carry look-ahead adder with its diagram. 15 3 3 2

- 27 (i) Evaluate $(A+B) * (C+D)$ using all 4 types of addressing instructions. (8 Marks) 15 4 2 2

- (ii) Discuss in detail the operational concepts of the computer system. (7 Marks)
